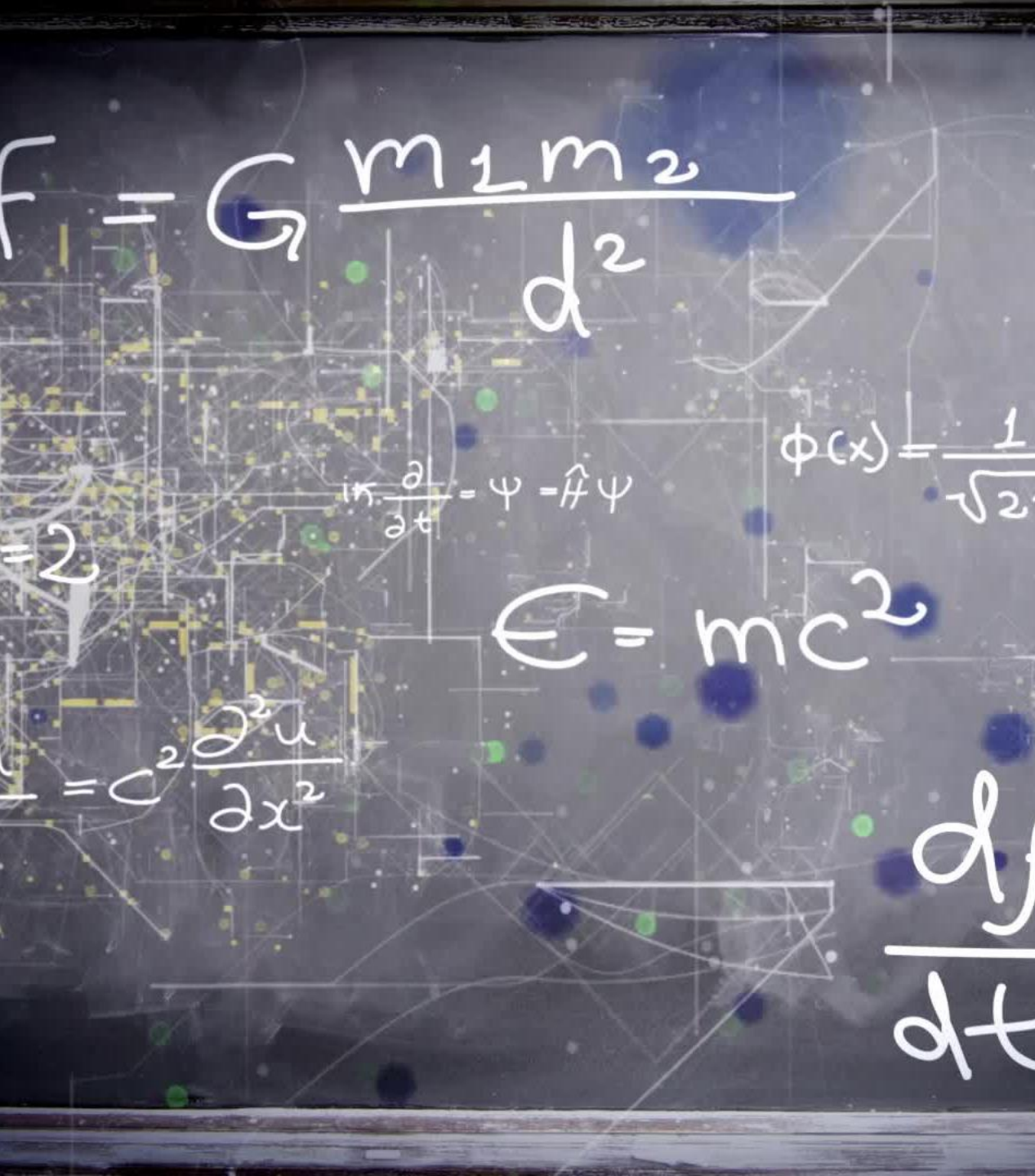




EOM232.03 – EXPLAIN THE CYCLES OF A FOUR-STROKE PISTON- POWERED ENGINE





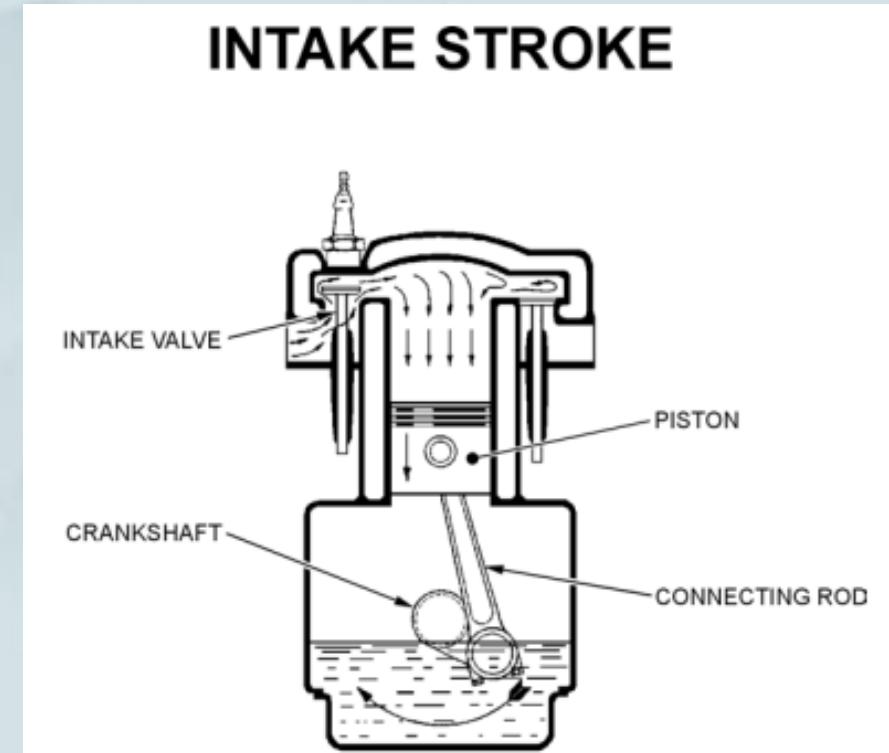
Teaching P0ints

- TP1 Explain the Purposes of the Strokes of a Four-stroke Piston
- TP2 Identify and Explain the Operation of Valves and Camshafts
- TP3 Explain the Timing of Electrical Ignition Spark Distribution
- TP4 Explain That all Engine Operations Must Proceed in Order



TP1 Explain the Purposes of the Strokes of a Four-stroke Piston

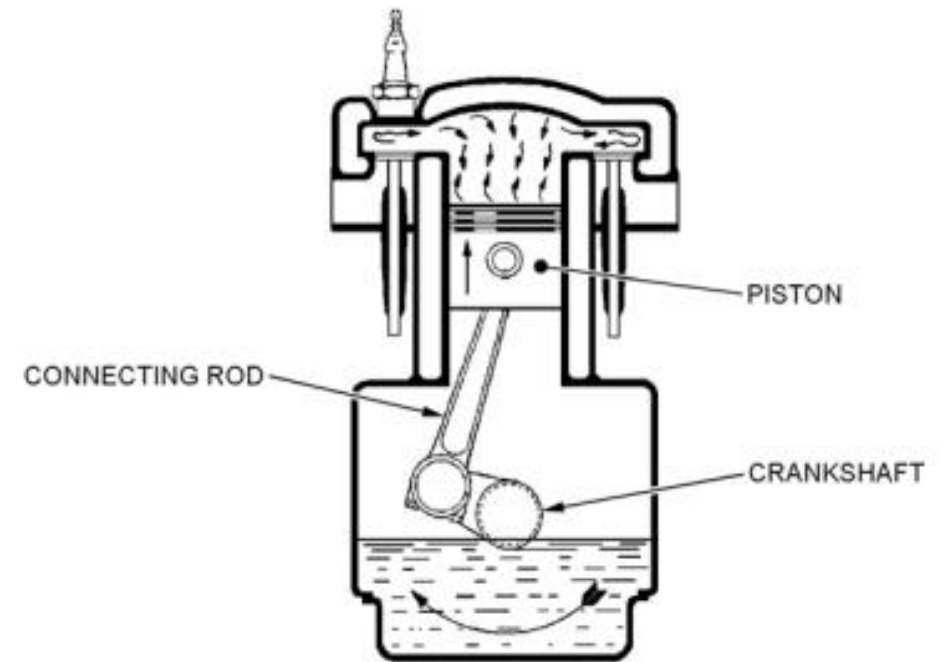
- The parts of an engine work together in a cycle to turn the aircraft's propeller. In most aircraft engines, this cycle has four distinct stages called strokes:



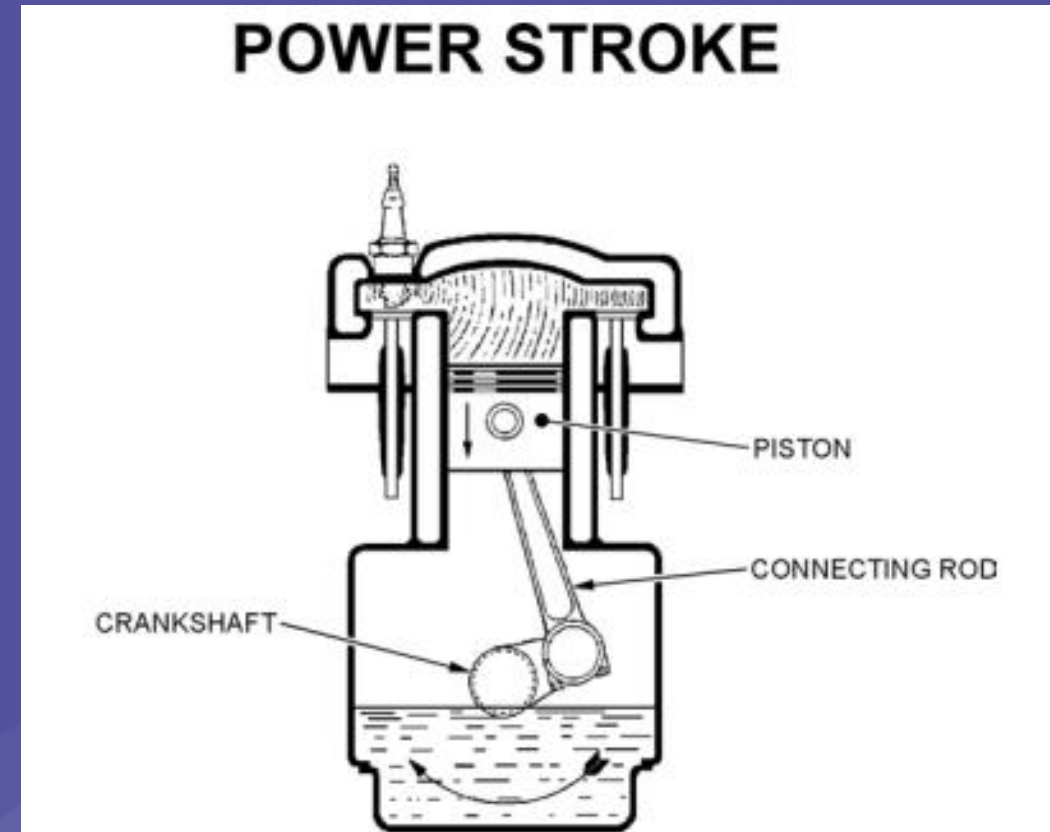
During the first (intake) stroke, the **intake valve opens** to let the gasoline and air mixture into the cylinder and the piston moves down to draw the mixture into the cylinder. The **exhaust valve is closed** during this stroke

- In the second (compression) stroke, **both valves are closed** while the piston moves up to compress the mixture

COMPRESSION STROKE

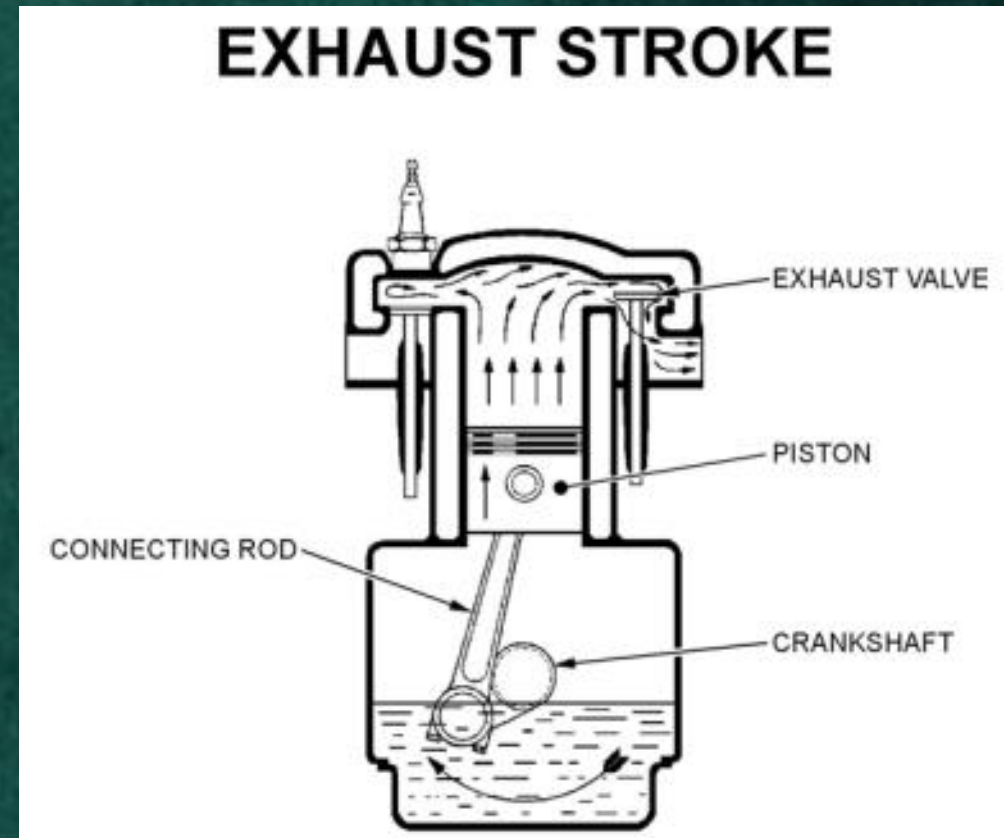


- In the third (power) stroke, both valves remain closed while the spark plug ignites the gas, which burns, expands and forces the piston down again.

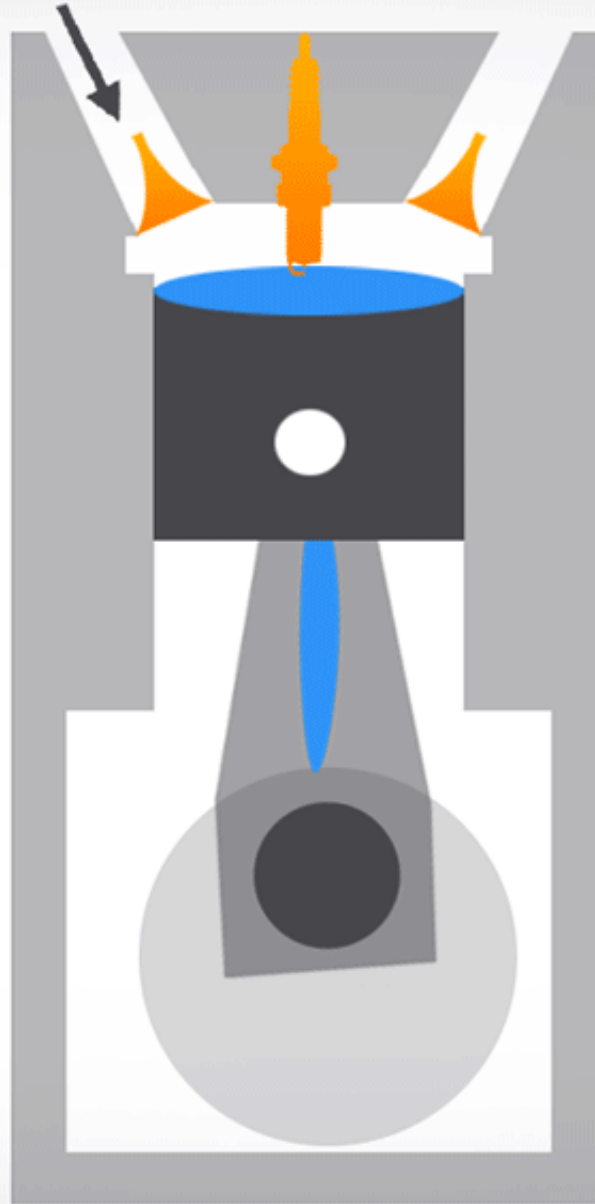


- In the fourth (exhaust) stroke, **the exhaust valve is open** to let the burnt gases out while the **intake valve is closed**. The piston moves up again to force the burned gases out through the open exhaust valve.

After the exhaust stroke, the whole process **repeats itself thousands of times per minute**, causing the crankshaft to turn the propeller on the aircraft.



Four Stroke Engine



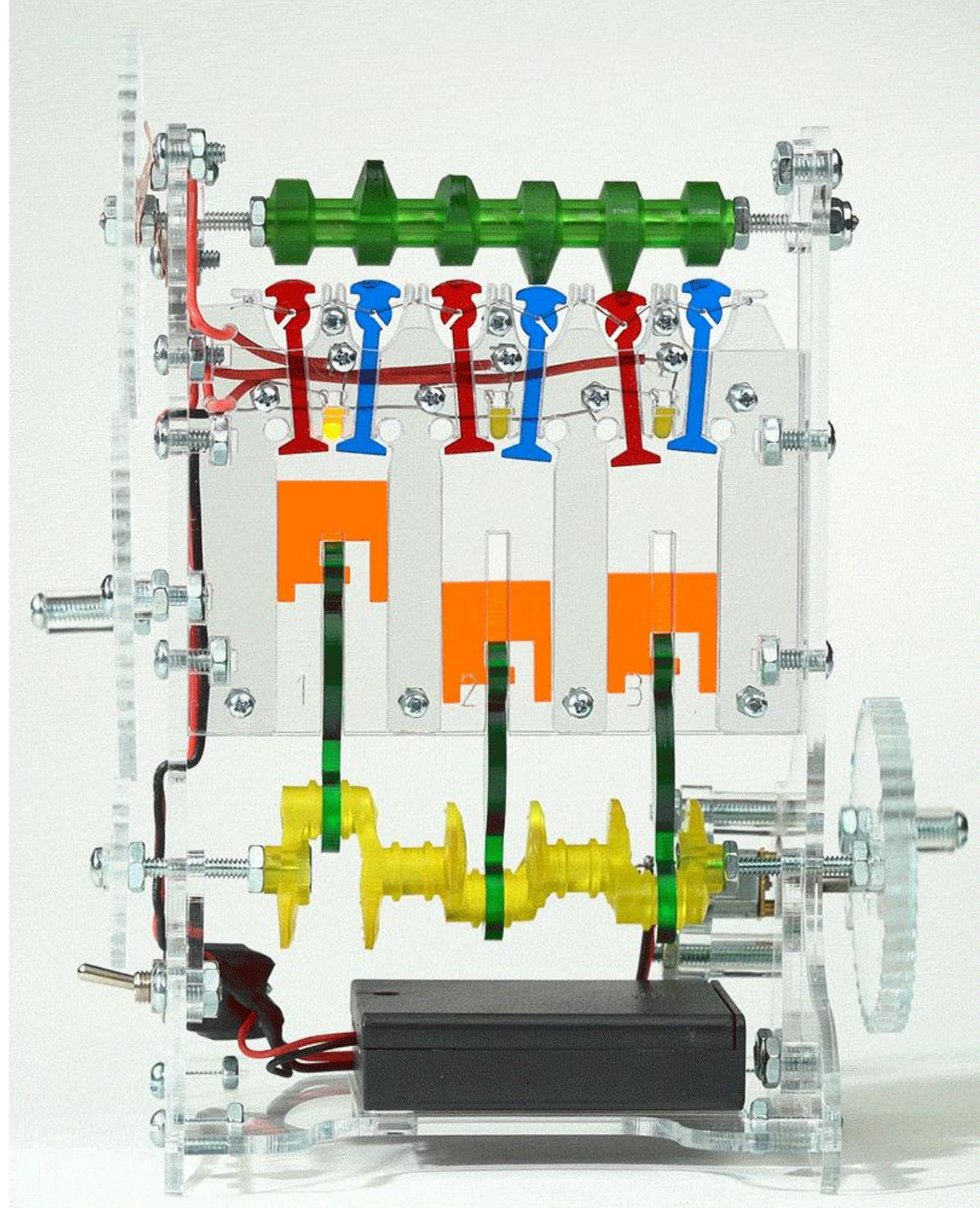
Simplified Engine

Stroke:
Intake



TP2 Identify and Explain the Operation of Valves and Camshafts

- Since the crankshaft rotates in time with the piston movements, **its rotation is used to provide signals to the valves**, telling them when to open.
- The crankshaft turns a secondary shaft (camshaft) that has cams, raised on its surface. The shape of the cam is such that **it mechanically pushes its associated valve open**



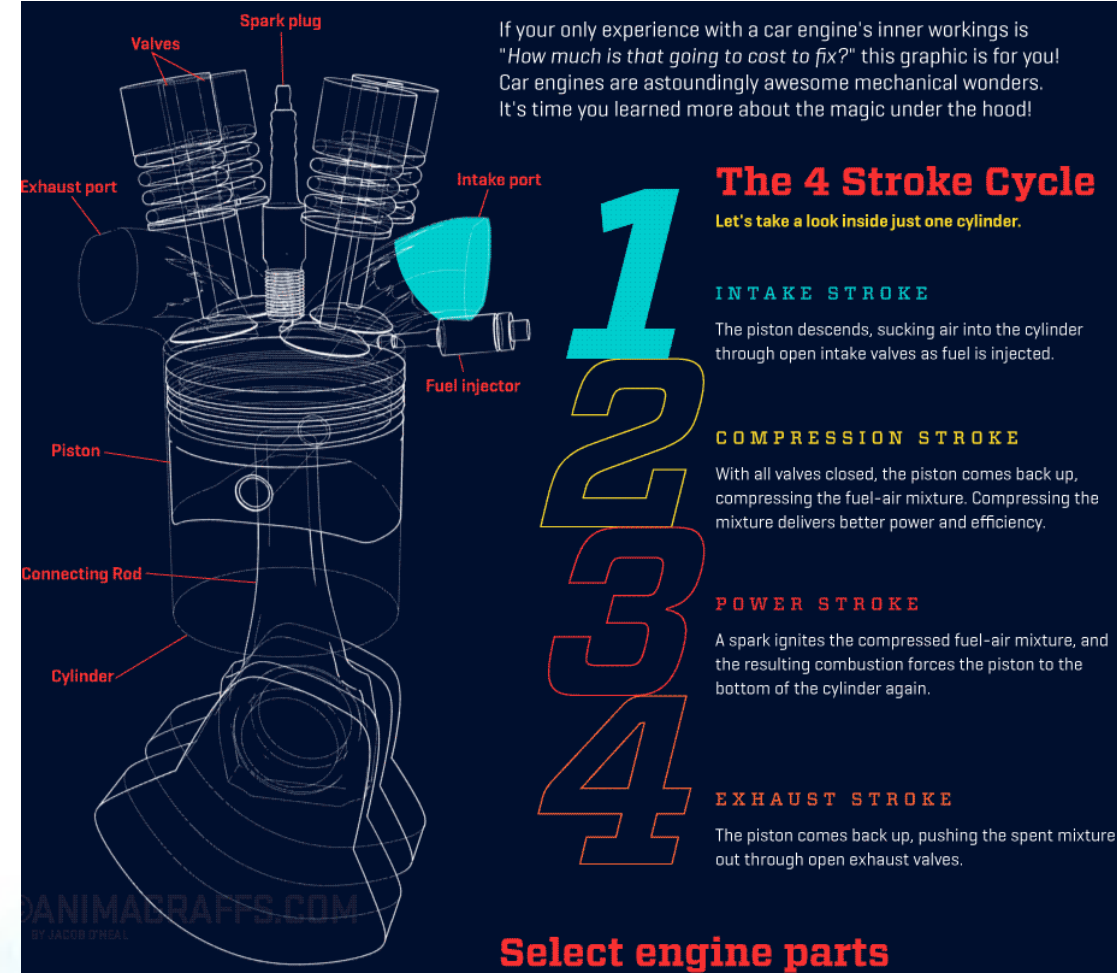
TP3 Explain the Timing of Electrical Ignition Spark Distribution

- Efficient, complete burning takes time. **Proper timing is required.**
- To ensure that the fuel is burned completely, and that all energy is recovered from the fuel, **the spark that sets off the detonation must be delivered while the piston is still rising on the compression stroke.**
- If the spark arrives during the power stroke there is not enough time to burn the fuel completely and unburned fuel is exhausted. This would be an inefficient waste of fuel Therefore, the timing of the spark plug's electrical signal must be exact.



- The engine operation must proceed as follows:
- Fuel and air mixture must be available for all cylinders, all the time, in a multi-cylinder engine
- The **intake stroke** of the piston (**intake valve open**)(**exhaust valve closed**)
- The **compression stroke** of the piston (both valves closed)
- Electrical signals must be delivered to **spark plugs** just before the piston completes the power stroke, when both valves are closed
- The **power stroke** of the piston (both valves closed)
- The **exhaust stroke** of the piston (**exhaust valve open**)(**intake valve closed**)
- The camshaft must push each valve open and closed at the right times

TP4 Explain That all Engine Operations Must Proceed in Order



VALVE MECHANISM

